NEERAJ VIJAYAKUMAR PATTANASHETTI

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Education

Master of Science | University of Illinois Chicago Aug, 2025 – May, 2027
Major in Computer Science **GPA: 4.0** (Expected Graduation: May, 2027) Chicago, USA

Bachelor of Engineering | Visveswaraya Technological University Dec, 2021 – Jul, 2025
Major in Computer Science **GPA: 3.52** Bengaluru, India

Work Experience

Software Engineering Intern | KreedaLoka
Jul, 2023 – Dec, 2023
Bengaluru, India

- Designed and deployed real-time application features for a competitive gaming platform using a **Java backend and a C#/.NET-based Unity client**, supporting **100+ concurrent users** with low-latency interactions.
- Optimized relational database schemas and SQL queries, **reducing data retrieval time by 40%** and ensuring consistent, real-time leaderboard updates under concurrent load.
- Implemented **tournament pairing logic** inspired by **Swiss-style systems**, initially seeding players by rating and dynamically pairing participants based on identical or near-identical scores in subsequent rounds (“winners play winners”), ensuring competitive balance as the tournament progressed.
- Implemented and scaled WebSocket-based real-time communication in a Java backend to deliver low-latency gameplay updates, state synchronization, and leaderboard refreshes while handling concurrent user connections reliably.

Projects

AI Conversational Avatar
Aug 2025

- Developed an **agentic** **AI-powered 3D conversational avatar** integrating Gemini LLMs to enable natural, voice-driven interactions, designed to simulate intelligent assistant behavior in real time.
- Built an agentic **Retrieval-Augmented Generation (RAG) pipeline** using Pinecone vector database to perform semantic context retrieval and dynamically guide model responses based on retrieved knowledge.
- **Implemented a full-stack architecture using React, Node.js, and RESTful APIs**, integrating speech-to-text and text-to-speech services to support real-time bidirectional voice workflows between the web frontend and AI backend.

PHI Shield - Healthcare AI Compliance System
Nov 2025

- Built **a** **healthcare-focused AI system** to detect and mitigate Patient Health Information (PHI) in clinical and enterprise data pipelines, aligning with HIPAA-compliant workflows.
- Architected and deployed real-time data pipelines using **Pathway** to monitor, transform, and analyze streaming and batch healthcare data, enabling continuous inspection of data flows for compliance, traceability, and downstream AI processing.
- Integrated **Aparavi with LLM-based PHI detection** and policy-driven enforcement, to prevent sensitive data exposure across structured and unstructured healthcare datasets in a HIPAA-aligned workflow.

AI-Driven Web Automation using Playwright & MCP
Oct 2024

- **Developed an** **AI-enhanced web automation solution** using **Playwright** and **Model Context Protocol (MCP)** to programmatically automate searches and data extraction from the Los Angeles Open Data Portal
- **Implemented full browser automation with Python**, integrating a **Playwright MCP server** and custom MCP client logic to control browser actions (navigate, fill, click, extract) reliably through structured page snapshots.
- **Built extensible system architecture** including an **AI agent planner (Gemini/LLM)**, Python backend, and optional **Flask REST API** with a UI to expose search functionality as a /search endpoint.

Skills

- **Programming Languages**: Python | Java | C# | JavaScript | HTML | CSS | Javascript | .NET / ASP.NET
- **Frameworks**: Node.js | Express | React.js | TensorFlow | PyTorch | REST APIs
- **AI / Machine Learning**: Machine Learning | Deep Learning | LLMs | RAG Pipelines | NLP | Computer Vision
- **Databases**: SQL | NoSQL (Firebase Realtime DB) | Data Modelling
- **Developer Tools & Platforms:** Git | Agile/Scrum | Jira | Playwright | Unity | OpenCV | Pathway | Aparavi

Awards & Achievements

Top 5 – HackWithChicago (Microsoft Chicago)

- Recognized among top teams for building **PHI Shield**, a production-grade, **AI-assisted PHI detection and redaction system.** Designed a **streaming data pipeline** to process sensitive healthcare text in real time, applied LLM-based analysis for entity detection, and focused on **system reliability, scalability, and real-world compliance use cases** rather than a demo-only solution.

Top 5 – Cursor AI Rapid Build Event

- Awarded for **idea quality and rapid execution** after building a voice-operated, OS-style system within a strict 1-hour time limit using **Cursor.** Implemented real-time voice input and carefully designed prompts to translate natural-language commands into structured system actions, demonstrating fast **prototyping and strong end-to-end execution** under pressure.